Regi Kusumaatmadja

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Education

VU Amsterdam and Tinbergen Institute, Ph.D. in Economics, Expected Completion: 2026.
Tinbergen Institute, M.Phil. in Economics, 2021.
VU Amsterdam, M.Sc. in Economics, 2018.
University of Indonesia, B.Sc. in Physics, 2016 (Best Graduate).

Letter Writers

prof. dr. José Luis Moraga-González
VU Amsterdam & Télécom Paris
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prof. dr. Sabien Dobbelaere
VU Amsterdam
Tinbergen Institute, IZA
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prof. dr. Mark J. Roberts
Pennsylvania State University
NBER, ZEW
Email: mroberts@psu.edu

prof. dr. Eric J. Bartelsman (placement dir.)
VU Amsterdam
Tinbergen Institute, IZA
Email: e.j.bartelsman@vu.nl

Visiting Position

2026.05 - 2026.06, CUNEF Universidad Madrid, invited by **dr. Costanza Cincotta**
2023.01 - 2023.06, the Pennsylvania State University, invited by **prof. dr. Mark J. Roberts**

Research and Teaching Areas

Primary Fields: Industrial Organization, Economics of Innovation
Secondary fields: Applied Econometrics, Applied Microeconomics

Research in Progress

Internal and External R&D: An analysis of costs and benefits

This paper analyzes the costs and benefits of in-house and contracted-out R&D among Dutch ICT firms from 2000 to 2020. I estimate a dynamic discrete-choice model with mode-specific investment costs. Contracted-out R&D costs about 2.45 million euros, versus 0.13 million for in-house R&D, which is why so few firms contract out alone. Doing both is cheaper than the sum of the two costs: in-house R&D offsets almost all of the extra cost of going outside. Productivity gains are similar across modes. A uniform subsidy raises R&D participation and firm value by more than a subsidy aimed only at in-house R&D.

Presentation: JEI - Gran Canaria (2022), Ph.D.-EVS (2022), IO/Applied Micro Brownbag - Penn State (2023), VU Amsterdam Internal Seminar (2023), EARIE - Rome (2023), Tinbergen Institute Ph.D. Seminar (2023), DKIIS - Valencia (2023), CAED - Penn State (2024), SED Meeting - Barcelona (2024), CEPR Workshop: Trade, Geography, and Industrial Organization - Erasmus Rotterdam (2024, poster), APIOC - Seoul (2024), CEPR Symposium - Paris (2024, poster), EEA-ESEM - Bordeaux (2025), CEPR IO Virtual Gathering (2025)

Entry into Innovation Areas

This paper investigates the multi-area entry decisions of listed U.S. electronics firms. I apply Latent Dirichlet Allocation to over 630,000 patents to define twenty innovation areas and merge them with market-implied valuations. I estimate a static entry game with moment inequalities. Firms enter areas where expected patent valuations are higher. Holding those valuations fixed, additional rivals reduce expected profit. Firms gain from occupying more areas at once. A large yearly per-area cost and rival congestion still limit expansion. From observed portfolios, a 25% reduction in that cost raises to 22–27% the share of firms that occupy an extra area.

Presentation: VU Amsterdam Internal Seminar (2025)

The Impact of Government Financial Assistance on the Scope and Direction of Firm Innovation, with C. Cincotta.

Governments spend hundreds of billions of dollars each year financing private innovation. We ask whether this spending expands, redirects, or reallocates research. A Salop model and stacked difference-in-differences on U.S. public firms from 2000 to 2021, combining Good Jobs First awards with patent data, compare recipients with close and distant rivals, using text-based product-market overlap. Distant rivals provide the preferred reading of the treated firm's own response. Recipients' patenting scale rises by about 3 percent relative to distant product-market controls. Close product-market rivals do not change how much they patent. Direction shifts toward the most-cited technologies, strongest for cash grants.

Mergers and Innovation: An Exploration of Scope and Direction of Innovation, with S. Dobbelaere and J. L. Moraga-Gonzalez.

How do mergers change the scale and composition of firm innovation? We study U.S. public mergers from 1980 to 2020 using combined acquirer–target patent portfolios. We measure *scope*, the scale of the joint portfolio, and *direction*, its alignment with highly cited patents. A compact framework isolates three mechanisms. Common ownership internalizes business-stealing externalities in product-market overlap areas and reduces the private return to innovation there. The merger may also pool technological assets, raising the productivity of R&D where the firms possess related knowledge. Some innovation lines require a fixed cost before they can be operated; pooled assets can make previously inactive lines worth activating. Product-market overlap and technological overlap are distinct: firms may compete for the same customers without sharing knowledge, or share related technologies without competing for the same customers. We compare treated merger pairs to matched control pairs before and after the deal. On the matched sample, joint-portfolio scope is larger after mergers than for matched controls. The difference is substantially larger when pre-merger technologies overlap, especially among non-horizontal deals. We find no strong evidence that activity is reallocated between technology classes shared by both firms and classes unique to one. Joint portfolios show higher alignment with highly cited patents relative to matched controls, although the framework leaves the sign of that change theoretically ambiguous. Acquirer-only R&D measures overstate post-merger input growth relative to joint-entity measurement.

Presentation: VU Amsterdam Internal Seminar (2024), Tinbergen Institute Workshop in Economics of Innovation (2025)

Teaching Experience

2021-2025 TA, VU Amsterdam, Introduction to Econometrics (MSc). Eval: 4.3/5.0
2021-2025 TA, VU Amsterdam, Applied Econometrics (MSc). Eval: 4.7/5.0
2022-2025 Supervisor, VU Amsterdam, Research Project and Thesis (BSc + MSc).
2021-2024 Instructor, VU Amsterdam, Math, Statistics, Micro, and R (Refresher MSc).
2022 TA, VU Amsterdam, Microeconomics I (BSc).
2021-2025 Guest Lecturer, VU Amsterdam, Urban Economic Challenges & Policies (MSc).

Relevant Experience

2023, EARIE 1st Summer School on Entry Model, LUISS - Rome.
2022, Econometric Society's Dynamic Structural Econometrics Summer School, MIT.
2019, Research on Productivity, Trade, and Growth Summer School, Tinbergen Institute.
2020-2021, RA for Eric J. Bartelsman and Sabien Dobbelaere, VU Amsterdam, MICROPROD project.
2018-2019, Research Associate, Institute for Economic & Social Research, University of Indonesia (LPEM FEB UI).

Scholarships, Honors, and Awards

2025, Excellence in Teaching Award, VU Amsterdam.
2023, Research Visit Grant, VU Amsterdam (monetary grant).
2019-2021, Tinbergen Institute - Full Scholarship (tuition & living cost).
2017, VU Amsterdam Fellowship Programme (tuition).
2017, Holland Scholarship Programme (living cost).
2016, Best Graduate, Department of Physics, University of Indonesia (monetary prize).
2016, Top 1 (one) Percent, Faculty of Mathematics and Natural Sciences, University of Indonesia.
2014-2016, XL-Axiata Future Leaders Scholarship (leadership programme & in-kind benefits).
2014-2015, Karya Salemba Empat Scholarship (tuition & living cost).

Publications

Evaluating the congestion-reducing effects of road rationing policy: Evidence from Jakarta's odd-even policy. *Case Studies on Transport Policy* (2025), with M. Halley Yudhistira, Y. Sofiyandi, and M. Firman Hidayat. (previously circulated as Does Traffic Management Matter? Evaluating Congestion Effect of Odd-Even Policy in Jakarta)

Theoretical exploration of optical response of Fe_3O_4-reduced graphene oxide nano-particle system within dynamical mean-field theory. *IOP Conf. Ser.: Mater. Sci. Eng.* (2017), with M.A. Majidi, A.D. Fauzi, W.Y. Phan, A. Taufik, R. Saleh, and A. Rusydi.

Policy Work

The sky in blues: on the recent development of Indonesian airlines industry. *LPEM FEB UI WP* (2019), with C. Nuryakin, N. W. G. Massiw, A. U. Lumbanraja, S. Hambali, and D. Anindita

Miscellaneous

Language: Indonesian (native), English (fluent), Dutch (basic-intermediate), Arabic (basic-intermediate)
Software: R, Python, Rust

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